Decision and Risk Analysis

Game Theory

DAEN 427/ISEN 427 Fall 2025

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From Parametric to Strategic Decisions

Parametric decisions: The outcome depends on your action and nature's state.
"Should I bring an umbrella?"
It depends on the weather, not what others do.

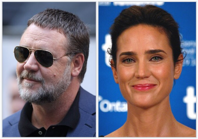
Strategic decisions: The outcome depends on your action *and* others' actions.
"Should I bring an umbrella when there's only one dry bus seat left?"

Strategic interaction means anticipating others' moves.
You're not just solving a problem, you're solving people.

- **Game theory** is the study of *mathematical models of strategic interactions* between rational decision-makers.
- Applications in many fields: economics, logic, computer science, behavioral science, biology.
- Modern game theory began with the work of John von Neumann and Oskar Morgenstern in their 1944 book *Theory of Games and Economic Behavior*.
- John Nash introduced the Nash equilibrium in the 1950s, extending the theory to non-zero-sum games.

von Neumann

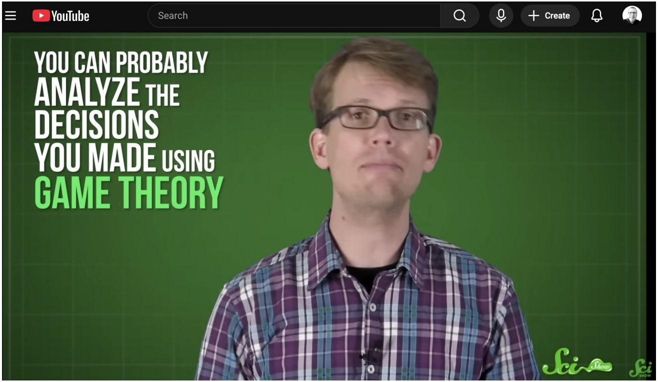
Nash



Russell Crowe (left) and Jennifer Connelly (right), play John and Alicia Nash, respectively.

A Beautiful Mind

A nice video (for your reference)

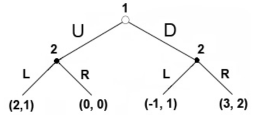


Key Types of Games

- **Cooperative vs Non-cooperative**
  – Cooperative: players can form binding agreements.
  – Non-cooperative: players act individually.
- **Symmetric vs Asymmetric**
  – Symmetric: players have identical strategy sets and payoffs only depend on strategies, not on player identity.
- **Zero-sum vs Non-zero-sum**
  – Zero-sum: one player's gain = another's loss.
  – Non-zero-sum: players may all gain or lose.
- **Simultaneous vs Sequential**
  – Simultaneous: players move without knowing other's choice.
  – Sequential: players move in turn, possibly observing earlier moves.

Representation of Games

- **Normal form (strategic form)**: payoff matrix showing each player's strategies & payoffs.
- **Extensive form**: decision tree showing sequential moves, information sets.
- **Characteristic function form**: used in cooperative games, describing value of each coalition.



Extensive Form

	(L,L)	(L,R)	(R,L)	(R,R)
U	2,1	2,1	0,0	0,0
D	-1,1	3,2	-1,1	3,2

Strategic Form

$$v(c) = \begin{cases} 80, & \text{if } c = \{A\} \\ 56, & \text{if } c = \{B\} \\ 70, & \text{if } c = \{C\} \\ 80, & \text{if } c = \{A, B\} \\ 85, & \text{if } c = \{A, C\} \\ 72, & \text{if } c = \{B, C\} \\ 90, & \text{if } c = \{A, B, C\} \end{cases}$$

Function Form

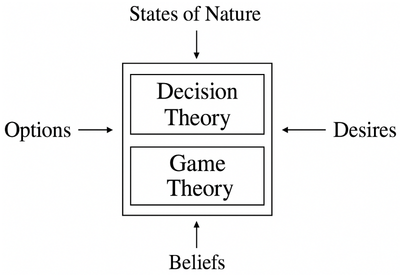
Applications

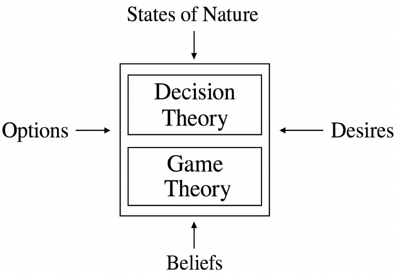
- **Economics:** oligopoly, auctions, bargaining.
- **Biology:** evolutionary strategies, competition.
- **Political science:** voting, war/peace strategies.
- **Social sciences:** negotiation, networks, behavioral.
- **Computer science / AI:** multi-agent systems, system design.

Although not always perceived as such, the theory is actually **multidisciplinary**.

Well-Known Examples

- **The Prisoner's Dilemma:** Illustrates why two rational individuals might not cooperate, even if it appears that it is in their best interest to do so.
- **The Battle of the Sexes:** Two players prefer to coordinate but have different preferences for coordination outcomes.
- **The Ultimatum Game:** A proposer offers a split of a sum; the responder accepts or rejects.





- Helps in understanding strategic behavior when outcomes depend on others' actions.
- Provides rigorous tools for modeling interaction in a wide range of fields.
- Highlights how rational decision-making can lead to suboptimal outcomes (e.g., Prisoner's Dilemma).
- Supports design of mechanisms, policies, or strategies where multiple decision-makers interact.

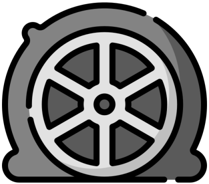
The Makeup Exam (A Classic Lie Gone Wrong)

Two students skip their exam and claim a flat tire.
Professor gives a makeup test.

Question 1 (5 points): Easy stats.
Question 2 (95 points): "Which tire?"

- Both write the same tire $\rightarrow$ both get A's.
- Different tires $\rightarrow$ both fail miserably.

This is a **coordination game**, the best outcome depends on matching the other player's answer. *Even lies require equilibrium.*



The Makeup Exam (A Classic Lie Gone Wrong)

Payoff Matrix:

	FL	FR	RL	RR
FL	A, A	F, F	F, F	F, F
FR	F, F	A, A	F, F	F, F
RL	F, F	F, F	A, A	F, F
RR	F, F	F, F	F, F	A, A

- Same tire $\longrightarrow$ Both get A's.
- Different tires $\longrightarrow$ Both fail.
- **Four Nash equilibria:** one for each tire.

The Prisoners' Dilemma

Two suspects. Two choices: Cooperate (C) or Defect (D).

Payoff Matrix (in years in jail):

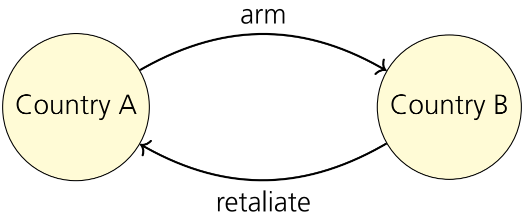
	C	D
C	(-2,-2)	(-20,0)
D	(0,-20)	(-10,-10)

Nash Equilibrium: Both defect. Because defection strictly dominates cooperation.

Lesson: Rational people can end up in irrational situations. Just like every group project ever.

In the Real World

- **Arms races:** Each country wants safety, ends up broke.
- **Overfishing:** Everyone wants"just one more fish."
- **Pollution:** Everyone drives, nobody breathes.
- **Corporate meetings:** Everyone talks, nothing gets done.



Individual rationality doesn't guarantee collective happiness.
Sometimes the"smart" move ruins the world (or the weekend).

Repeated Games: Hope for Humanity?

If the game repeats indefinitely, cooperation can emerge.
Why? Because players can *punish* defectors in future rounds.

Example: "You didn't do the dishes yesterday—guess who's sleeping on the couch tonight?"

But if there's a last round, backward induction kills cooperation:

- Last round $\rightarrow$ no reason to cooperate.
- Therefore, second to last $\rightarrow$ no reason either.

Lesson: The only way to cooperate forever is to never end the game.

Nash Equilibrium ("No Regrets" Rule)

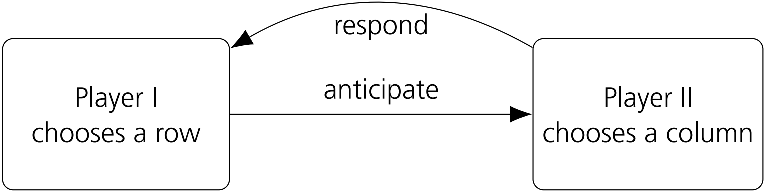
- A **Nash equilibrium** is a concept in game theory where no player can improve their outcome by changing their own strategy, assuming all other players keep their strategies unchanged.
- In simpler terms: everyone is doing the best they can given what others are doing. **No one has an incentive to deviate.**

  Example: In the Prisoner's Dilemma, both players choosing to "confess" is a Nash equilibrium — neither can get a better outcome by unilaterally switching to "stay silent."

  So, a Nash equilibrium represents a stable state where all players' decisions are mutually consistent.

Nash Equilibrium ("No Regrets" Rule)

Definition: Each player's strategy is the best response to others'. No one can improve by unilaterally changing.



Example: In the flat tire, both write "Front Left."
Neither can improve $\rightarrow$ equilibrium.

Nash Equilibrium (The Logic of "No Regrets")

Definition: A Nash equilibrium is a strategy profile where each player's choice is the best response to the others' choices.

In plain English:"Given what you're doing, I can't do better by changing."

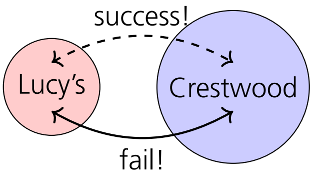
Example: In the flat tire, if both write"Front Left," neither gains by switching. $\Rightarrow$ Each is doing the best they can given the other's (bad) lie.

Moral: Equilibrium $\neq$ best outcome. It just means nobody can complain *without being a hypocrite*.

Coffee Shop Coordination

You and your friend forgot to pick which café, Lucy's or Crestwood.

	Lucy's	Crestwood
Lucy's	(1,1)	(0,0)
Crestwood	(0,0)	(1,1)



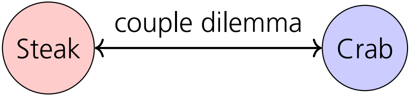
Two pure Nash equilibria:

- (Lucy's, Lucy's)
- (Crestwood, Crestwood)

Moral: Coordination is easy if you have telepathy or text messages (unless you're trying to text each other from the subway).

The Battle of the Sexes (aka Netflix Negotiations)

A couple wants to dine together but have different preferences.



Payoffs:

	Steak House	Crab Shack
Steak House	(2,1)	(0,0)
Crab Shack	(0,0)	(1,2)

Equilibria:

- (Steak House, Steak House)
- (Crab Shack, Crab Shack)

Analogy: Like when one wants to watch"John Wick" and the other"Bridgerton." Either way, someone's pretending to enjoy it.
Lesson: Rational harmony sometimes means pretending to enjoy the other's choice.

When Pure Strategies Fail

Sometimes there's no equilibrium in pure strategies. Enter the **mixed strategy**.

Example: Matching Pennies (or Coffee Shops Redux)
You want to avoid your ex; your ex wants to find you.
No pure equilibrium, but a mixed one exists: both randomize 50/50.

Translation: If you can't predict them, confuse them. (Also known as"dating.")

Battle of the Sexes in Mixed Strategies

In mixed equilibrium:

- Player I goes to Steak House with probability $p = 2/3$.
- Player II goes to Crab Shack with probability $q = 1/3$.

Both are indifferent between choices.

Takeaway: You can't win every time—but you can confuse your partner enough that no one knows who's losing.

Spousonomics (Game Theory at Home)

Book premise: "Economics is the route to marital bliss."
Example: Randomize chores.

- Sometimes do the laundry.
- Sometimes don't.
- Keep your partner guessing.

Result: Higher household efficiency. Possibly higher divorce risk.
Mixed strategies: not just for poker—also for surviving marriage.

Trembling-Hand Perfection

Even rational players slip up (literally tremble). A strategy is **trembling-hand perfect** if it still works when others might make tiny mistakes ($\varepsilon > 0$).

Example: You"accidentally" like your ex's 2017 photo. If your strategy survives that tremble, it's perfect.

Key: Robust equilibria survive human clumsiness.

Subgame Perfection (Planning Every Move)

Sequential games have multiple stages. You must think ahead and reason backward (**backward induction**).

Example: Parent and child:

- Child:"If I cry, I get candy."
- Parent:"If I give candy now, I'll regret it later."

Backward induction says: think from the end. **Moral:** Rational parenting = consistent punishment and lots of snacks.

MAD – Mutually Assured Destruction

Nuclear strategy:"If you bomb me, we all die."

It's a Nash equilibrium, but not **subgame perfect** because the threat isn't credible.

Solution: Dr. Strangelove's doomsday machine, automatic retaliation!

Lesson: To make a threat credible, automate it … or just build something so terrifying that no one calls your bluff.

The Stag Hunt – Trust and Coordination

Two hunters can chase a deer (big reward) or hunt hares (small reward).

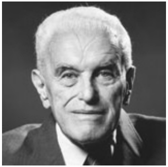
	Deer	Hare
Deer	(3,3)	(0,1)
Hare	(1,0)	(1,1)

Two equilibria:

- Both cooperate for deer $\rightarrow$ high reward.
- Both settle for hares $\rightarrow$ safe but boring.

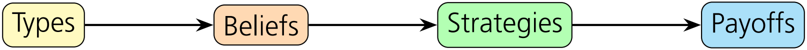
Moral: Societies thrive when people trust others to hunt the deer.

From Incomplete Information to Bayesian Games



- Introduced by John C. Harsanyi (1967–68).
- A **Bayesian game** models strategic interaction with private information.
- Each player has a **type** that represents hidden knowledge.
- Players have **beliefs** (probability distributions) about others' types.

Flow Diagram

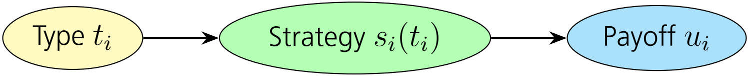


Structure of a Bayesian Game

A Bayesian game consists of:

- Players $i = 1, 2, \ldots, n$
- Action sets A_i
- Type sets T_i
- Common prior $p(t_1, t_2, \ldots, t_n)$
- Payoffs $u_i(t_1, \ldots, t_n, a_1, \ldots, a_n)$

Dependency Schematic



Strategies and Beliefs

- Each player chooses a strategy function $s_i : T_i \to A_i$ that maps types to actions.
- Players maximize expected utility based on their beliefs.

Example: Auction

- Each bidder's valuation (type) is private.
- Strategy: bid as a function of valuation.

Bayesian Nash Equilibrium (BNE)

A **Bayesian Nash Equilibrium** is a strategy profile $(s_1^*, \ldots, s_n^*)$ such that for every player i and every type t_i,

$$s_i^*(t_i) \in \arg\max_{a_i \in A_i} \mathbb{E}_{t_{-i}}[u_i(a_i, s_{-i}^*(t_{-i}), t_i, t_{-i})].$$

- Each player's strategy maximizes expected payoff given beliefs and others' strategies.
- Generalizes Nash equilibrium to settings with private information.

Optimization Flowchart

Beliefs $p(t_{-i})$ $\longrightarrow$ Expected Payoff $\longrightarrow$ Best Response s_i^*

Homework Assignment

Due: Tue Nov 20 9:35 AM

Read $\rightarrow$ answer the **blue** questions!

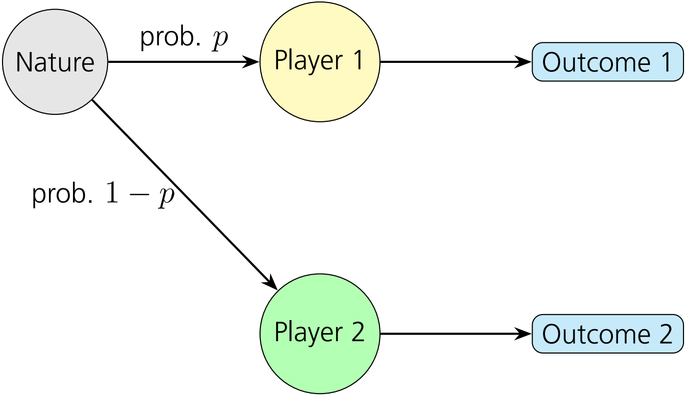
Use bullet points or full sentences.

Stochastic Outcomes

- Many games include **random events** that affect payoffs (e.g., dice rolls, market shocks).
- These are represented using a random variable or a special **“Nature” player**.
- A stochastic game may involve:
    - Randomly determined states or transitions.
    - Probabilistic payoffs based on chance.
    - Belief updates when new information is revealed.
- Players maximize their **expected utility** across all possible outcomes.

Stochastic Outcomes

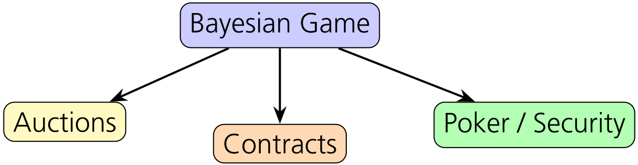
Nature's Move Diagram



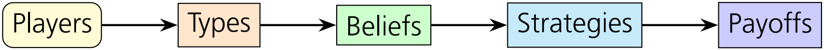
Applications of Bayesian Games

- **Auctions:** Players bid based on private valuations.
- **Contract theory:** Principal–agent problems with hidden information.
- **Adversarial settings:** Poker, security resource allocation.
- **Market competition:** Firms with private cost data.

Concept Map



- Game theory analyzes strategic interactions among rational players.
- Bayesian games model incomplete information using types and beliefs.
- Players choose strategies based on expected utilities.
- The key equilibrium concept: **Bayesian Nash Equilibrium**.



The World as a Game

Game theory teaches:

- People don't act alone, they anticipate others.
- Nash equilibria = stable, but not necessarily happy.
- Cooperation is fragile but possible (or, not impossible :)
  Remember: Every group chat, meeting, and marriage is secretly a repeated Prisoners' Dilemma.

In real life:

- Rational ≠ sensible.
- Mixed strategies are underrated.
- And every social interaction is a low-stakes prisoners' dilemma.

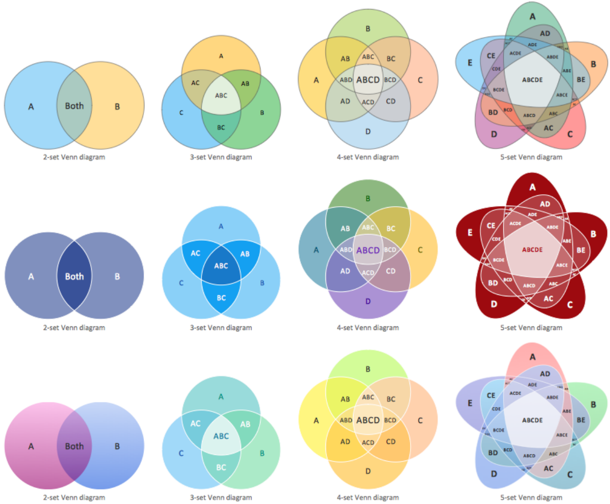
“Life is a series of coordination failures disguised as social events.”

Main takeaway:

- Know the game.
- Anticipate others’ moves.
- And if you lie about a flat tire, agree on <u>which one.</u>

One More Thing ...

- 7 sets Venn Diagram
- Venn Diagrams



Evolutionary dynamics of any multiplayer game on regular graphs